

# Humberto Sandmann



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Computer scientist and educator with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. With more than a decade of university teaching alongside senior industry practice in software architecture, machine learning, and high-performance computing, I bring research depth and applied engineering experience into the classroom.

## Research Interests

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DeFi, Smart contracts and decentralized applications

Small-world networks and complex systems

Computation in large-scale systems

Artificial neural networks and deep learning

Nonlinear systems and network dynamics

Synchronization of coupled oscillators

## Education

|                                                           |                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">mar.2012</a><br>↑                             | D.Sc.<br>thesis                                                                     | <a href="#">Polytechnic School of the University of Sao Paulo</a><br><a href="#">Spike patterns and computation in dynamical neural networks</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">mar.2007</a>                                  | doi<br>advisor<br>presentation<br>supported by                                      | <a href="#">10.11606/T.3.2012.tde-05092012-165022</a><br><a href="#">Prof. Dr. Emilio Del Moral Hernandez</a><br><a href="#">slides</a><br><a href="#">CNPq - grant 140632/2007-7</a><br>( <a href="#">mar.2007</a> → <a href="#">dec.2009</a> ) ∧ ( <a href="#">sep.2010</a> → <a href="#">feb.2011</a> )                                                                                                                                                                                                                                                                                                                                                                         |
| <a href="#">sep.2010</a><br>↑<br><a href="#">aug.2009</a> | D.Sc. Internship<br>advisor<br>supported by                                         | <a href="#">Max Planck Gesellschaft für Dynamik und Selbstorganisation</a><br><a href="#">Prof. Dr. Marc Timme</a><br><a href="#">CAPES - grant BEX 4076-09-5</a><br><a href="#">jan.2010</a> → <a href="#">aug.2010</a><br><br><a href="#">Pró-Reitoria da Pós-Graduação da USP</a><br><a href="#">aug.2009</a> ∧ <a href="#">dec.2009</a> (flight tickets GRU ⇌ FRA)<br><br><a href="#">José Humberto Sandmann - my father</a><br><a href="#">aug.2009</a> → <a href="#">dec.2009</a> (German language classes/exams)<br><br><a href="#">Max Planck Institute - Networks Dynamics Group</a><br><a href="#">aug.2009</a> → <a href="#">nov.2009</a> (accommodation in Goettingen) |
| <a href="#">jun.2006</a><br>↑                             | M.Sc.<br>dissertation                                                               | <a href="#">Polytechnic School of the University of Sao Paulo</a><br><a href="#">Prediction of time series using architecture based on neuro-fuzzy systems</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <a href="#">feb.2003</a>                                  | doi<br>advisor                                                                      | <a href="#">10.11606/D.3.2006.tde-01042009-095125</a><br><a href="#">Prof. Dr. Marco Túlio Carvalho de Andrade</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <a href="#">dec.2002</a><br>↑<br><a href="#">feb.1999</a> | B.Sc.                                                                               | <a href="#">Faculdade de Engenharia Industrial</a><br><a href="#">Computer Science Bachelor</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <a href="#">dec.1998</a><br>↑<br><a href="#">feb.1996</a> | <a href="#">Escola Técnica Jorge Street</a><br>Technician in Industrial Computation |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



## Academic Appointments & Teaching

|                      |                                             |                                                   |                                                                                                                                                                           |                                                 |
|----------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| now<br>↑<br>jun.2018 | <a href="#">Insper</a><br>Sao Paulo, Brazil |                                                   |                                                                                                                                                                           |                                                 |
|                      | now<br>↑<br>feb.2020                        | Associate Professor                               |                                                                                                                                                                           |                                                 |
|                      |                                             | <a href="#">Platform, Microservices and APIs</a>  | <a href="#">Spring Cloud, Spring Boot, PostgreSQL, Redis, RabbitMQ, Kafka, Docker, K8s</a>                                                                                | 2026.1, 2025.2, 2025.1, 2024.1                  |
|                      |                                             | <a href="#">Coding Interviews</a>                 | Course to prepare for coding interviews.                                                                                                                                  | 2026.1                                          |
|                      |                                             | Machine Learning                                  | PCA, t-SNE, UMAP, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Classification, Regression, Clustering, EDA, Pre-processing, Feature Engineering | <a href="#">2026.1</a>                          |
|                      |                                             | <a href="#">Neural Networks and Deep Learning</a> | Artificial Neural Networks, Perceptron, Multi Layer Perceptron, Convolutional Neural Networks, GAN, VAE, CLIP, Stable Diffusion                                           | 2025.2                                          |
|                      |                                             | TecWeb                                            | JS, React                                                                                                                                                                 | <a href="#">2025.2</a>                          |
|                      |                                             | <a href="#">Cloud Computing</a>                   | <a href="#">MAAS, Juju, OpenStack, K8s, IaaS, PaaS, Computer Networks</a>                                                                                                 | 2025.1, 2024.2                                  |
|                      |                                             | <a href="#">Design of Software</a>                | Python                                                                                                                                                                    | 2025.0, 2024.1, 2023.2, 2022.1, 2021.2, 2021.1  |
|                      |                                             | Information System for Administration and Economy | Excel, VBA, Python, NumPy, Pandas, Stochastic Processes, RoI                                                                                                              | 2023.2, 2023.1, 2022.2                          |
|                      |                                             | Spreadsheet Automation in Excel                   | Excel, VBA                                                                                                                                                                | <a href="#">2021.1</a> , <a href="#">2020.1</a> |
|                      | jan.2020<br>↑<br>aug.2018                   | Assistant Professor                               |                                                                                                                                                                           |                                                 |
|                      |                                             | <a href="#">Design of Software</a>                | Python                                                                                                                                                                    | 2019.2, 2019.1                                  |
|                      |                                             | Foundations for Physical Communication            | Python, Fourier Transform, Wavelet Transform, Signal Processing                                                                                                           | 2018.2                                          |

|          |                                                          |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
|----------|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| dec.2025 | <a href="#">ESPM</a>                                     |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
| ↑        | Sao Paulo, Brazil                                        |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
| jun.2014 | Associate Professor                                      |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
|          | <a href="#">Machine Learning</a>                         | Machine Learning, PCA, KNN, K-Means, Decision Trees, Random Forests, Support Vector Machines, Naïve Bayes, Page Rank, PySpark, Classification, Regression, Clustering, Pre-processing, Feature Engineering                                                | 2025.2                                                                                                                                                                                                                 |
|          | <a href="#">Applied Mathematics</a>                      | Propositional Logic, Boolean Algebra, Set Theory, Polynomials, Functions, Sequences, Series, Logarithm, Combinatorial Analysis, Limits, Derivatives, Linear Algebra, Matrix Algebra, Graph Theory                                                         | <a href="#">2025.1</a> , <a href="#">2024.1</a> , <a href="#">2023.2</a> , <a href="#">2023.1</a> , <a href="#">2022.2</a> , <a href="#">2022.1</a> , <a href="#">2021.2</a> , <a href="#">2021.1</a> , 2020.2, 2020.1 |
|          | <a href="#">Oriented Object Programming</a>              | Java, Design Patterns, Docker, Spring Cloud, Microservices                                                                                                                                                                                                | 2025.1, 2024.1, 2023.1, 2022.1, 2021.1                                                                                                                                                                                 |
|          | Statistics                                               | Statistics, Probability, Random Variables, Distributions, Sampling, Estimation, Hypothesis Testing, Regression Analysis                                                                                                                                   | <a href="#">2023.2</a>                                                                                                                                                                                                 |
|          | Software Quality                                         | Software Quality, Software Testing, Software Architecture, Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation, Continuous Integration, Continuous Delivery | 2024.2, <a href="#">2022.2</a>                                                                                                                                                                                         |
|          | Architecture and Operating Systems                       | Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems                                                       | <a href="#">ESPM 16-bits</a><br>2021.1, 2020.2, 2020.1, 2019.2, 2019.1, 2018.2, 2018.1, 2017.1, 2016.2, 2016.1, 2015.2, 2015.1, 2014.2                                                                                 |
|          | Artificial Intelligence and Society                      | Artificial Intelligence, Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs, Transformers, LSTMs, BERT, Ethics, Bias                                                                 | 2022.1 (IR), <a href="#">2021.1</a> , <a href="#">2020.2</a> , <a href="#">2020.1</a> , <a href="#">2019.2</a>                                                                                                         |
|          | <a href="#">Introduction for Artificial Intelligence</a> | Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, PCA, KNN, K-Means, GANs                                                                                                                                   | <a href="#">2020.2</a> , <a href="#">2020.1</a> , <a href="#">2019.2</a> , <a href="#">2019.1</a>                                                                                                                      |
| apr.2015 | <a href="#">Toledo Prudente Centro Universitário</a>     |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
| ↑        | Presidente Prudente, Brazil                              |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
| mar.2015 | Visiting Professor                                       |                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                        |
|          | <a href="#">MBA: Big Data and Machine Learning</a>       | Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr, R, Octave, Self-Organizing Maps, Bayesian Decision Trees, Multilayer Perceptron                                                                                                                      | 2015.1                                                                                                                                                                                                                 |

|          |                                              |                                                                                                                                                                                                                                              |  |                   |
|----------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------|
| dec.2015 | <a href="#">FIAP</a>                         |                                                                                                                                                                                                                                              |  |                   |
| ↑        | Sao Paulo, Brazil                            |                                                                                                                                                                                                                                              |  |                   |
| mar.2012 | Associate Professor                          |                                                                                                                                                                                                                                              |  |                   |
|          | Distributed systems                          | Sockets, RMI, SOAP, Microservices, J2EE, Design Patterns                                                                                                                                                                                     |  | 2015              |
|          | MBA: Big Data and No-SQL Modelling           | Big Data, Hadoop, Hive, HBase, MongoDB, Neo4j, Solr                                                                                                                                                                                          |  | 2014.2            |
|          | Data Structure and Algorithm II              | C, Data Structure, Algorithm, Complexity, Linked List, Stack, Queue, Tree, Red-Black Tree, AVL Tree, B-Tree, Hash Table, Graph, Dijkstra Algorithm, Prim Algorithm                                                                           |  | 2013              |
|          | Algorithms                                   | C, Programming, Algorithm, Complexity                                                                                                                                                                                                        |  | 2013,<br>2012.2   |
|          | Database I and II                            | SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Object-Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence, Synonym |  | 2013,<br>2012     |
|          | Programming III                              | Pointers, C, C++, Object-Oriented Programming, Inheritance, Polymorphism, Encapsulation, Abstract Class, Interface, Exception Handling                                                                                                       |  | 2012              |
|          |                                              |                                                                                                                                                                                                                                              |  |                   |
| jul.2013 | <a href="#">Universidade Cruzeiro do Sul</a> |                                                                                                                                                                                                                                              |  |                   |
| ↑        | Sao Paulo, Brazil                            |                                                                                                                                                                                                                                              |  |                   |
| mar.2012 | Associate Professor                          |                                                                                                                                                                                                                                              |  |                   |
|          | Mobile Programming                           | Android                                                                                                                                                                                                                                      |  | 2012.2            |
|          | Computer Architecture                        | Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems                                          |  | 2012.1            |
|          | Software Engineering                         | Software Engineering, Software Development Life Cycle, Agile Methodologies, Test-Driven Development, Behavior-Driven Development, Test Automation                                                                                            |  | 2012.1            |
|          | Introduction to Programming                  | C, Programming, Algorithm                                                                                                                                                                                                                    |  | 2012.1            |
|          | Database II                                  | SQL, PL/SQL, Oracle, Transaction, Concurrency Control, Locking, Deadlock, Recovery, Normalization, ER Model, Relational Model, Trigger, View, Index, Stored Procedure, Function, Package, Cursor, Sequence                                   |  | 2013.1,<br>2012.1 |

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|----------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| jul.2011 | <a href="#">Universidade Nove de Julho</a>                        |                                                                                                                                                                                                        |                 |
| ↑        | Sao Paulo, Brazil                                                 |                                                                                                                                                                                                        |                 |
| mar.2011 | Temporary Professor                                               |                                                                                                                                                                                                        |                 |
|          | Computer Architecture                                             | Computer Architecture, Operating Systems, Binary and Hexadecimal Number Systems, Number Representation, Data Representation, Assembly Language, Memory Management, Process Management, File Systems    | 2011.1          |
|          | <a href="#">Introduction to Programming</a>                       | C, Programming, Algorithm                                                                                                                                                                              | 2011.1          |
|          | <a href="#">Web Programming</a>                                   | HTML, CSS, JavaScript                                                                                                                                                                                  | 2011.1          |
| jul.2011 | <a href="#">Polytechnic School of the University of Sao Paulo</a> |                                                                                                                                                                                                        |                 |
| ↑        | Sao Paulo, Brazil                                                 |                                                                                                                                                                                                        |                 |
| mar.2008 | Teaching Assistant                                                |                                                                                                                                                                                                        |                 |
|          | Experimental Pattern Recognition, Modeling and Neurocomputing     | Logic Gates, Linear Regression, Pattern Classification, <a href="#">UC Irvine Machine Learning Repository</a> , <a href="#">NIST Data Gateway</a> , Multilayer Perceptron, EEG Signals, PCA            | 2011.1*         |
|          | Modeling for Signal Processing: Neural Networks and Learning      | Artificial Neural Networks, Multilayer Perceptron, Backpropagation, Forecasting, Time Series, Classification                                                                                           | 2009.1, 2008.1* |
|          | Laboratory of the Foundations of Computing Engineering            | Computer Architecture, Assembly Language, Memory Management, Process Management, Register, Stack, Turing Machine, Finite State Machines, Von Neumann Architecture, Java, Orientated Object Programming | 2008.1†         |
|          | * grants from CAPES (PAE)                                         |                                                                                                                                                                                                        |                 |
|          | † grants from FAPESP                                              |                                                                                                                                                                                                        |                 |

## Professional Experience

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|          |                                                                         |                                                    |
|----------|-------------------------------------------------------------------------|----------------------------------------------------|
| mar.2025 | H-Sandmann                                                              | Machine Learning, Software Architecture.           |
| ↑        | Sao Paulo Area, Brazil                                                  |                                                    |
| feb.2006 | Founder                                                                 |                                                    |
| oct.2023 | SelsanTech                                                              | Machine Learning, Software Architecture.           |
| ↑        | Sao Paulo Area, Brazil                                                  |                                                    |
| mar.2012 | Co-Founder                                                              |                                                    |
| jun.2019 | RockSpoon                                                               | POS, ERP, Payroll, JavaEE, Postgres, Software      |
| ↑        | Sao Paulo Area, Brazil - Palo Alto, United States                       | Architecture, OpenCV, Machine Learning,            |
| may.2018 | Senior Software Architect and Machine Learning Engineer                 | Computer Vision, GoLang, Android, Kotlin.          |
| jul.2016 | Banco Votorantim                                                        | RoE, Banking Business, High-Performance            |
| ↑        | Sao Paulo Area, Brazil                                                  | Computing, J2EE, Spring Boot, Automating           |
| oct.2015 | Contractor: Senior Consultant for Computational Optimization            | Testing, Jenkins.                                  |
| nov.2011 | TecBan                                                                  | ATM, J2EE, InMemory Database, High-                |
| ↑        | Sao Paulo Area, Brazil                                                  | Performance Computing.                             |
| aug.2011 | Contractor: Senior Consultant for Computational Optimization            |                                                    |
| jun.2007 | ABN Amro Bank                                                           | Stock Market, Trading, Core Banking, Software      |
| ↑        | Sao Paulo, Brazil                                                       | Architecture, Cryptography, HSM, BouncyCastle,     |
| feb.2006 | Contractor: Senior Software Engineer/Architect                          | PKI, PKCS, Java, Servlet, Database.                |
| jan.2006 | Politec                                                                 | J2EE, Software Quality, Software Architecture,     |
| ↑        | Sao Paulo, Brazil                                                       | Banking Business.                                  |
| aug.2005 | Senior System Analyst                                                   |                                                    |
| jul.2005 | EverSystems                                                             | PKI, BouncyCastle, Cryptography, HSM, Core         |
| ↑        | Sao Paulo, Brazil                                                       | Banking, Stock Market, Java, Servlet, Database,    |
| jul.2004 | Senior Software Engineer/Architect - BankBoston - CitiBank - Elly Lilly | Communication Protocol, Synchronization and        |
|          |                                                                         | Replication.                                       |
| mar.2003 | Central Online                                                          | Object-Oriented Programming, <a href="#">LMS</a> . |
| ↑        | Sao Paulo, Brazil                                                       |                                                    |
| sep.2002 | Senior System Analyst                                                   |                                                    |
| sep.2002 | Datamace                                                                | Payroll, ERP, Software Architecture.               |
| ↑        | Sao Paulo Area, Brazil                                                  |                                                    |
| oct.1999 | Senior System Analyst                                                   |                                                    |
| oct.1999 | Heraeus Electro-Nite                                                    | Compliance, ERP, PCP.                              |
| ↑        | Sao Paulo Area, Brazil                                                  |                                                    |
| mar.1998 | Computer Programmer                                                     |                                                    |

## Publications

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Selected peer-reviewed publications. Full and up-to-date list: [ORCID 0000-0001-7758-695X](#).

E. Del-Moral-Hernandez; H. Sandmann; G. Araujo; **Context dependent pattern recognition - A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information**. IEEE Transactions on Neural Networks, 2009. doi: [10.1109/IJCNN.2009.5179061](#).

L.A. da Silva; H. Sandmann; E. Del-Moral-Hernandez; **A self-organizing architecture of recursive elements for continuous learning**. IEEE Transactions on Neural Networks, 2008. doi: [10.1109/IJCNN.2008.4634190](#).

E. Del-Moral-Hernandez; H. Sandmann; L.A. da Silva; **Pattern recovery in networks of recursive processing elements with continuous learning**. IEEE Transactions on Neural Networks, 2004. doi: [10.1109/IJCNN.2004.1379877](#).

## Grants & Funding

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|                    |                                                                         |                                                  |
|--------------------|-------------------------------------------------------------------------|--------------------------------------------------|
| <b>CNPq</b>        | Doctoral fellowship — grant 140632/2007-7                               | mar.2007 → dec.2009 $\wedge$ sep.2010 → feb.2011 |
| <b>CAPES</b>       | Doctoral internship abroad (Max Planck Institute) — grant BEX 4076-09-5 | jan.2010 → aug.2010                              |
| <b>CAPES (PAE)</b> | Teaching assistantship fellowship — Polytechnic School, USP             | 2008, 2011                                       |
| <b>FAPESP</b>      | Teaching assistantship fellowship — Polytechnic School, USP             | 2008                                             |



## Academic Service & Memberships

### **IEEE** – Institute of Electrical and Electronics Engineers

Member #60216475

- **Computational Intelligence Society (CIS)** – member since 2004; Brazilian committee member (apr.2009 → aug.2009 ^ oct.2010 → mar.2012)
- **Signal Processing Society (SPS)** – 2005 → 2009
- **Engineering in Medicine and Biology Society (EMBS)** – 2005 → 2009

### **INNS** – International Neural Network Society

Student member #3951

## Languages

|                   |           |
|-------------------|-----------|
| <b>Portuguese</b> | native    |
| <b>English</b>    | advanced  |
| <b>German</b>     | <b>B1</b> |